



Foo Inc. Business Continuity Plan

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RISK ASSESSMENT REPORT

FOO INC.

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Risk Assessment

Background:

Foo, Inc. is a business in the video game industry. This business currently employs 40 employees and is known for their many successful video games in the industry. The founders have focused on first person, action-adventure, and battle royale game genres. Foo, Inc. is responsible for creating known franchises such as Grand Theft Robbery Series, Fortnite 2, and Star Wares. Many of these games are intended to be played online or in a multiplayer aspect. The company distributes their products through various ways like app stores, console games, and through PC gaming on Steam and Epic Games.

Purpose:

This assessment will focus on three parts for the evaluation. Firstly, the document will cover the **risk assessment, defining threats and vulnerabilities**, and provide scores based on their impacts. Secondly, plans for controlling these risks will be provided below for protecting intellectual property and patch management. Finally, this report will display recovery and incident response within contingency planning.

Objectives:

This report's objective is to identify threats in an organized document. By compiling relevant information, this report aims to provide understanding of various online threats. The information in this document is crucial for mitigating risks and enhancing security protocols. Foo, Inc's risk assessment is a valuable resource for protecting digital assets and maintaining recent practices.

Assumptions:

The following assumptions were made for the purpose of this report:

- Assessment focuses on protecting Foo, Inc's online games and assets
- Threats may be network-based, and target online infrastructure
- Company applies standard security practices (encryption and authentication)

Scenario

The Company

Risk Assessment

This initial risk assessment was conducted in accordance with the NIST SP 800-30 framework, focusing on the organizational Tier (Tier 1). Risks were identified by analyzing potential threat events, their likelihood of occurrence, existing system vulnerabilities, current mitigation controls, and the potential impact on organization operations and mission.

Table 1: Threat Sources (Information based on NIST SP)

Type of Threat Source	Description	Characteristics
ADVERSARIAL - Individual (outsider, insider, trusted, privileged) - Group (ad-hoc or established) - Organization (competitor, supplier, partner, customer) - Nation state	Individuals, groups, organizations, or states that seek to exploit the organization's dependence on cyber resources (i.e., information in electronic form, information and communications technologies, and the communications and information-handling capabilities provided by those technologies).	Capability, Intent, Targeting
ACCIDENTAL - User - Privileged User/Administrator	Erroneous actions taken by individuals while executing their everyday responsibilities	Range of effects
STRUCTURAL - Information Technology (IT) Equipment - Storage - Processing - Communications - Display - Sensor - Controller - Environmental Controls - Temperature/Humidity Controls - Power Supply - Software - Operating System - Networking - General-Purpose Application - Mission-Specific Application	Failures of equipment, environmental controls, or software due to aging, resource depletion, or other circumstances which exceed expected operating parameters.	Range of effects

The tables below are based on the NIST SP 800-30 to assess how likely a threat event is likely to occur.

Table 2: Assessment Scale Likelihood of Threat (Adversarial)

Qualitative Score (Description)	Semi-Quantitative Values	Description	
Very High	96-100	10	Adversary is almost certain to initiate the threat event
High	80-95	8	Adversary is highly likely to initiate the threat event
Moderate	21-79	5	Adversary is somewhat likely to initiate the threat event
Low	5-20	2	Adversary is unlikely to initiate the threat event
Very Low	0-4	0	Adversary is highly unlikely to initiate the threat event

Table 3 Assessment Scale – Likelihood of Threat Occurrence (Non-Adversarial)

Qualitative Values	Semi-Quantitative Values	Description
Very High	96-100	Error, accident, or act of nature is almost certain to occur, or occurs more than 100 times per year.
High	80-95	Error, accident, or act of nature is highly likely to occur, or occurs between 10-100 times per year.
Moderate	21-79	Error, accident, or act of nature is somewhat likely to occur; or occurs between 1-10 times per year.
Low	5-20	Error, accident, or act of nature is unlikely to occur; or occurs less than once a year, but more than once every 10 years
Very Low	0-4	Error, accident, or act of nature is highly unlikely to occur, or occurs less than once every 10 years.

Table 4: Assessment Scale – Impact of Threat Events

Qualitative Values (Description)		Semi-Quantitative Values	Description
Very High	96-100	10	The threat event could be expected to have multiple severe or catastrophic adverse effects on organizational operations, organizational assets, individuals, other organizations, or the Nation.
High	80-95	8	The threat event could be expected to have a severe or catastrophic adverse effect on organizational operations, organizational assets, individuals, other organizations, or the Nation. A severe or catastrophic adverse effect means that, for example, the threat event might: (i) cause a severe degradation in or loss of mission capability to an extent and duration that the organization is not able to perform one or more of its primary functions; (ii) result in major damage to organizational assets; (iii) result in major financial loss; or (iv) result in severe or catastrophic harm to individuals involving loss of life or serious life threatening injuries
Moderate	21-79	5	The threat event could be expected to have a serious adverse effect on organizational operations, organizational assets, individual's other organizations, or the Nation. A serious adverse effect means that, for example, the threat event might: (i) cause a significant degradation in mission capability to an extent and duration that the organization is able to perform its primary functions, but the effectiveness of the functions is significantly reduced; (ii) result in significant damage to organizational assets; (iii) result in significant financial loss; or (iv) result in significant harm to individuals that does not involve loss of life or serious life threatening injuries.
Low	5-20	2	The threat event could be expected to have a limited adverse effect on organizational operations, organizational assets, individual's other organizations, or the Nation. A limited adverse effect means that, for example, the threat event might: (i) cause a degradation in mission capability to an extent and duration that the organization is able to perform its primary functions, but the effectiveness of the functions

			is noticeably reduced; (ii) result in minor damage to organizational assets; (iii) result in minor financial loss; or (iv) result in minor harm to individuals.
Very Low	0-4	0	The threat event could be expected to have a negligible adverse effect on organizational operations, organizational assets, individual's other organizations, or the Nation.

Table 5: Assessment Scale – Level of Risk

Qualitative Values	Semi-Quantitative Values		Description
Very High	96-100	10	Threat event could be expected to have multiple severe or catastrophic adverse effects on organizational operations, organizational assets, individuals, other organizations, or the Nation.
High	80-85	8	Threat event could be expected to have a severe or catastrophic adverse effect on organizational operations, organizational assets, individuals, other organizations, or the Nation.
Moderate	21-79	5	Threat event could be expected to have a serious adverse effect on organizational operations, organizational assets, individuals, other organizations, or the Nation
Low	5-20	2	Threat event could be expected to have a limited adverse effect on organizational operations, organizational assets, individuals, other organizations, or the Nation
Very Low	0-4	0	Threat event could be expected to have a negligible adverse effect on organizational operations, organizational assets, individuals, other organizations, or the Nation.

Table 6: Assessment Scale – Level of Risk (Combination of Likelihood and Impact)

Likelihood (That occurrence results in adverse impact)	Level of Impact				
	Very Low	Low	Moderate	High	Very High
Very High	Very Low	Low	Moderate	High	Very High
High	Very Low	Low	Moderate	High	Very High
Moderate	Very Low	Low	Moderate	Moderate	High
Low	Very Low	Low	Low	Low	Moderate
Very Low	Very Low	Very Low	Very Low	Low	Low

Risk assessment results

Threat Event	Threat Source (vulnerability)	Mitigating Factors	Overall Likelihood (Table 2 or 3)	Level of Impact (Table 4)	Risk (Tables 5 and 6)
DDOS ATTACKS Conduct simple Denial of Service (DoS) attack. (Tier 3) Conduct Distributed Denial of Service. (Tier 3) (DDoS) attacks.	Cause: Lack of traffic tools and network authentication	Deploy Firewalls and an intrusion prevention system. Enable rate limiting to limit the number of requests from a single IP	High	Low	Low

Conduct targeted Denial of Service (DoS) attacks. (Tier 2)		Monitor, log, and analyze traffic in real time			
Social Engineering Compromise organizational information systems to facilitate exfiltration of data/information. (Tier 3) Obtain sensitive information via exfiltration. (Tier 3) Cause integrity loss by injecting false but believable data into organizational information systems. (Tier 3)	Cause: Exploiting human error Phishing: False emails/messages created to mimic legitimate sources, aiming to steal sensitive information like passwords, emails, and other information Spear phishing: Phishing attacks aimed at high value targets like executives Creating counterfeit/Spoof websites: False websites that mimic legitimate sites, gathering sensitive information or downloading malware.	Training and tracking awareness of phishing and false websites. Regular testing of local false phishing attempts, providing training if failure.	High	High	High
Advanced Persistent Threats Exploit recently discovered	Cause: outdated systems or unpatched systems.	Frequent patching of software, frequent updates.	Moderate	Low	Low

vulnerabilities. (Tier 3)		Monitoring of vulnerabilities.			
Network Intrusion Conduct attacks using unauthorized ports, protocols and services. (Tier 3) Conduct externally based network traffic modification (man-in-the-middle attacks). (Tier 3) Conduct wireless jamming attacks. (Tier 3)	Cause: Weak encryption and weak communication through signals.	Firewalls for ports blocking unusual connections. Remote Access: use a virtual private network with strong encryption. Conduct regular audits of user accounts. Deploy firewalls and update firmware routinely.	Low	Very High	Moderate

References

Joint Task Force Transformation Initiative. (2012, September). *Guide for Conducting Risk Assessments*. NIST National Institute of Standards and Technology.
<https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-30r1.pdf>.

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